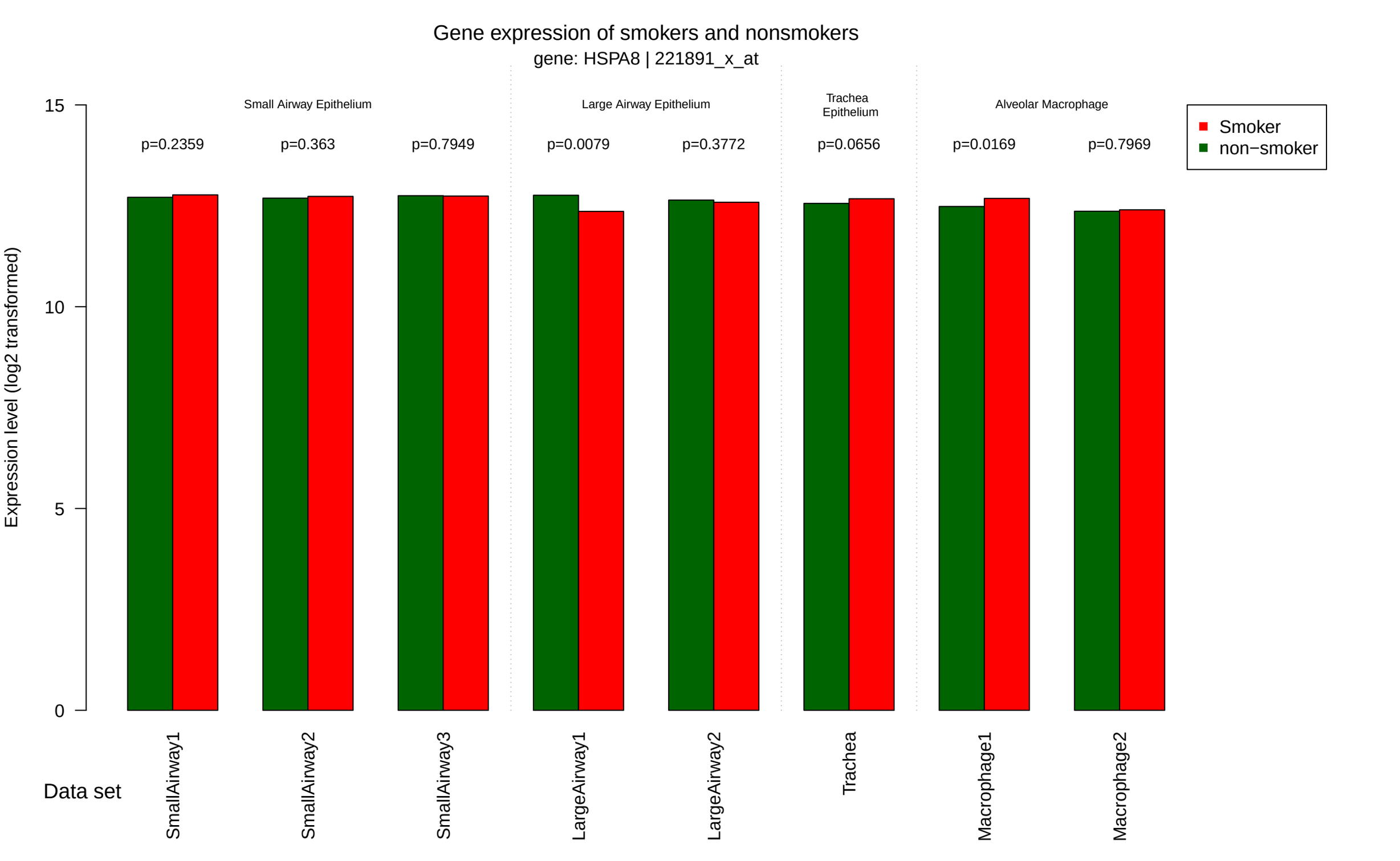
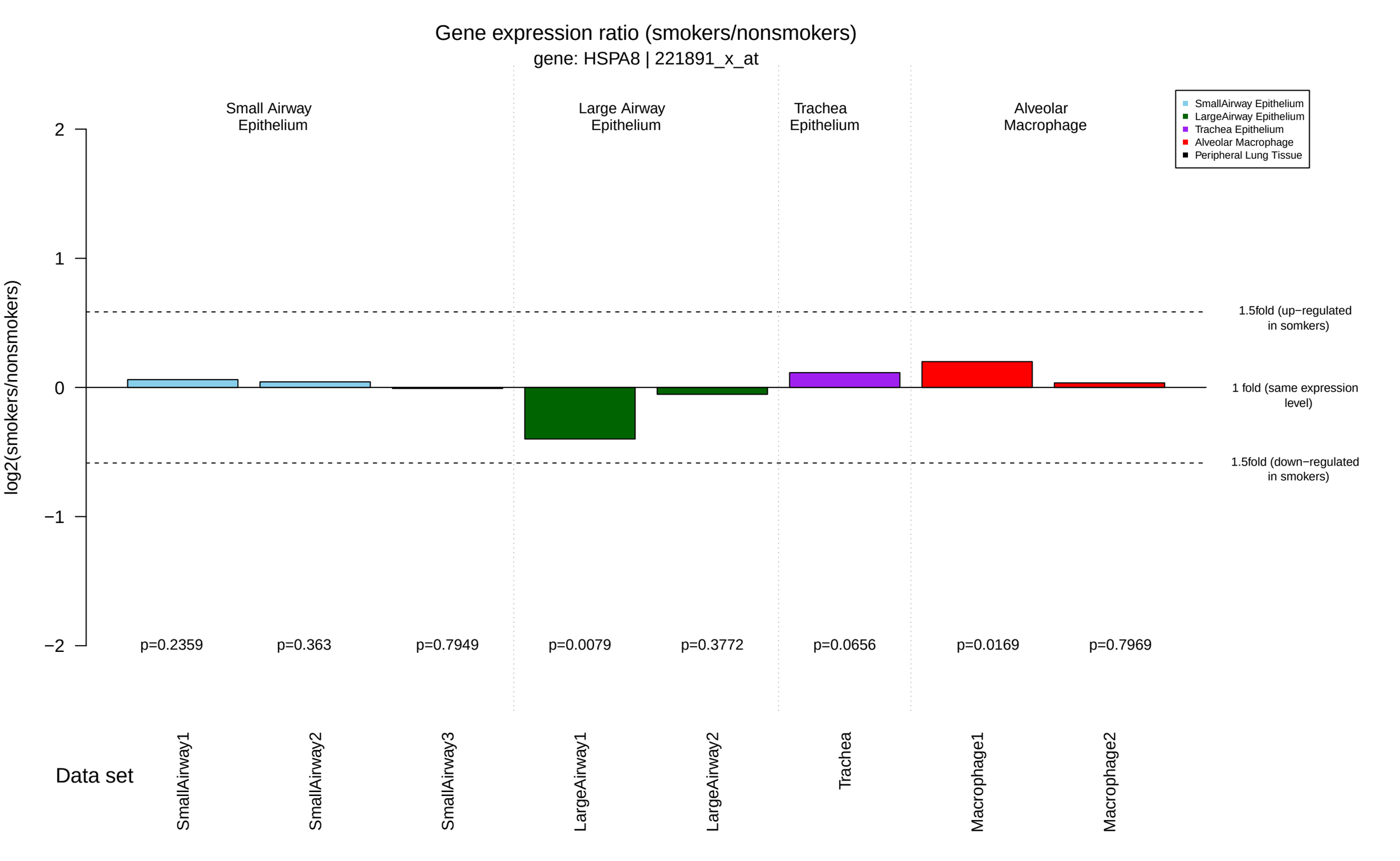
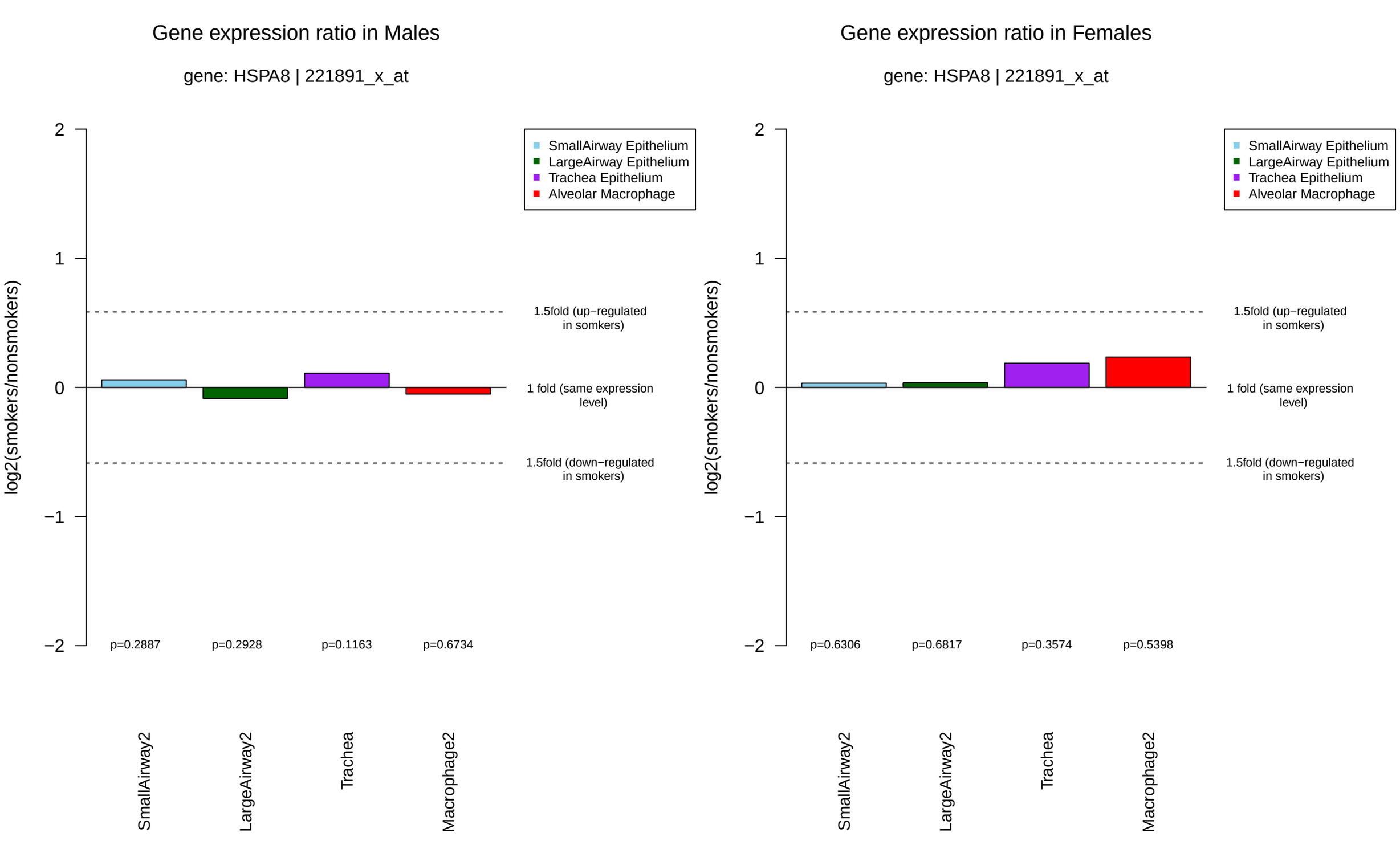
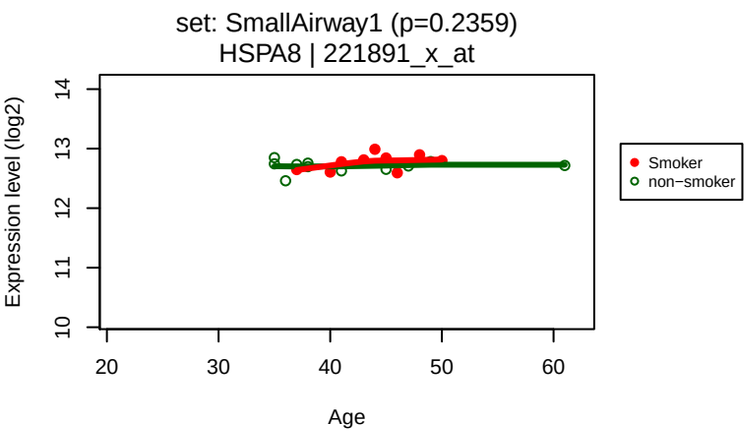
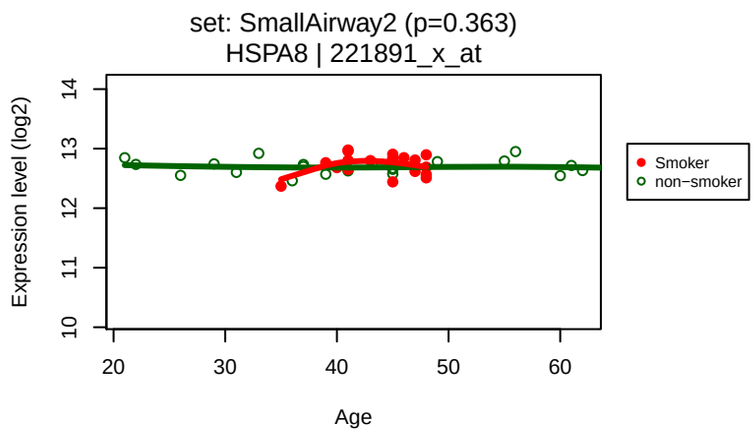
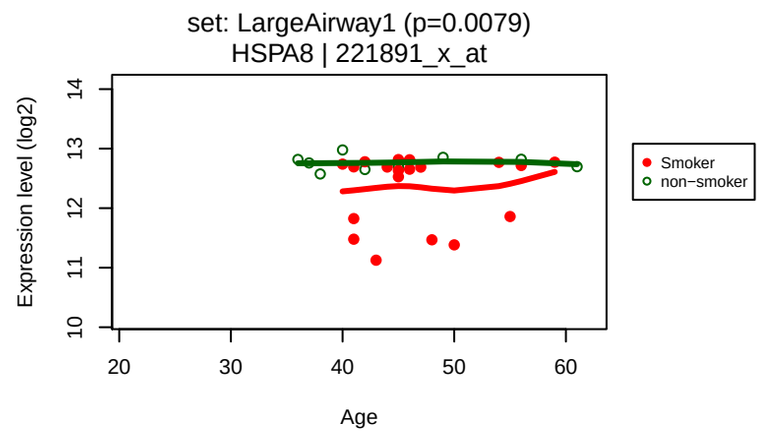

Gene expression of smokers and nonsmokers
gene: HSPA8 | 221891_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2359
p=0.363
p=0.7949
p=0.0079
p=0.3772
p=0.0656
p=0.0169
p=0.7969
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

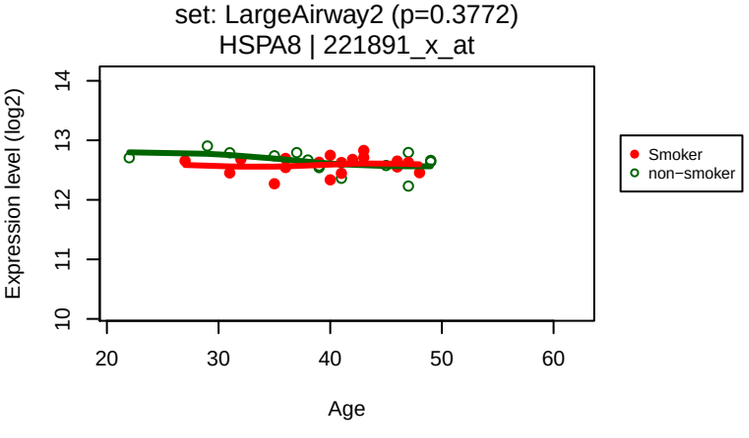

Gene expression ratio (smokers/nonsmokers)
gene: HSPA8 | 221891_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2359
p=0.363
p=0.7949
p=0.0079
p=0.3772
p=0.0656
p=0.0169
p=0.7969
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

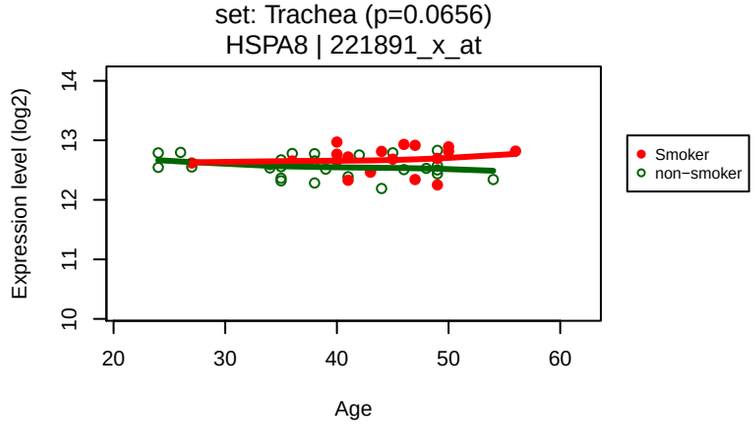

Gene expression ratio in Males
gene: HSPA8 | 221891_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2887
p=0.2928
p=0.1163
p=0.6734
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HSPA8 | 221891_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6306
p=0.6817
p=0.3574
p=0.5398
SmallAirway2
LargeAirway2
Trachea
Macrophage2

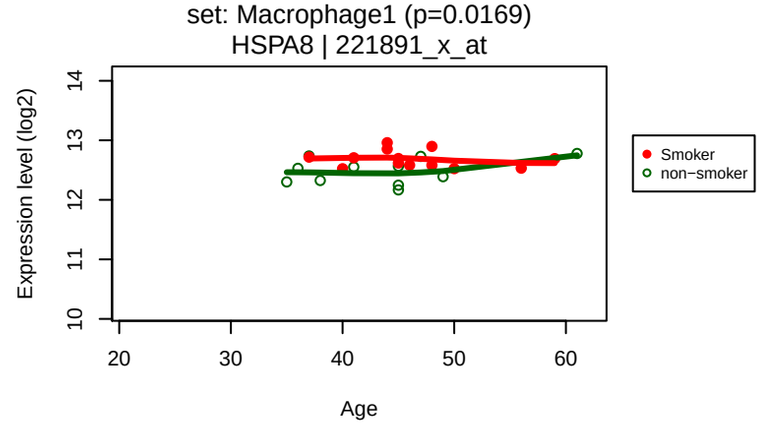

set: SmallAirway1 (p=0.2359)
HSPA8 | 221891_x_at
Expression level (log2)
Age
Smoker
non−smoker

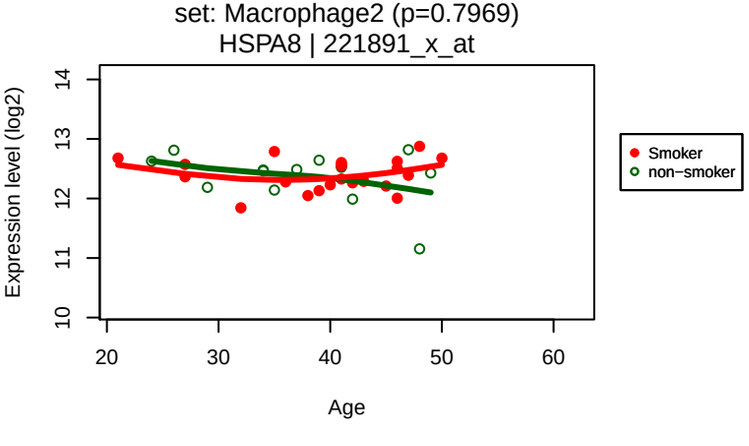

set: SmallAirway2 (p=0.363)
HSPA8 | 221891_x_at
Expression level (log2)
Age
Smoker
non−smoker

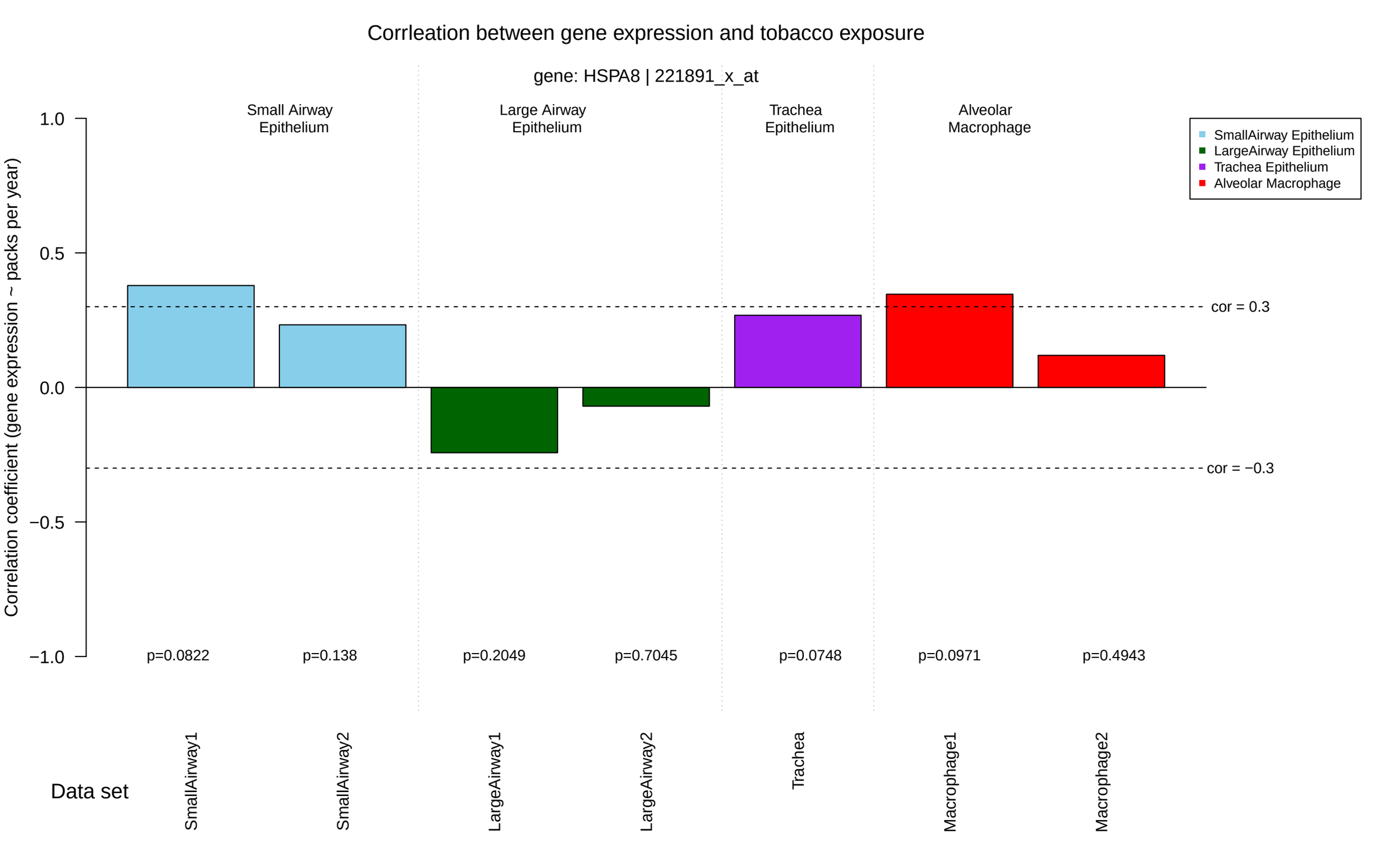

set: LargeAirway1 (p=0.0079)
HSPA8 | 221891_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3772)
HSPA8 | 221891_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0656)
HSPA8 | 221891_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0169)
HSPA8 | 221891_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7969)
HSPA8 | 221891_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HSPA8 | 221891_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0822
p=0.138
p=0.2049
p=0.7045
p=0.0748
p=0.0971
p=0.4943
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2